UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/904,768	10/02/2024	Yu Fang	SONA-0006-U01-C01-C01	1990

87084 7590 03/25/2026

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER
KASSA, ELIZABETH

ART UNIT	PAPER NUMBER
2457	

NOTIFICATION DATE	DELIVERY MODE
03/25/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/904,768	Applicant(s) Fang et al.	
	Examiner ELIZABETH KASSA	Art Unit 2457	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/02/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-22 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/02/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 01/07/2025, 01/07/2025.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Claims 1- 22 have been presented for examination and are rejected.

Information Disclosure Statement

3. The information disclosure statements (IDS) submitted on 01/07/2025 and 01/07/2025. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Double Patenting

4. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*,

686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

5. Claims 1-22 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-29 of U.S. Patent No. US 11252039 hereinafter '039. Although the claims at issue are not identical, they are not patentably distinct from each other because each element of the claims of the instant application is anticipated claim 1 of patent '039.

Similar reasoning applies to claim 14 of the instant application. See, the table below which shows both application claims on limitation bases.

The instant application - 18/904,768	Patent No. US 11252039
Claim 1. A system, comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description; a configuration circuit structured to configure a first network interface circuit in response to the network regulation description; and wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.	Claim 1. A system, comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description; and a configuration circuit structured to configure a first network interface circuit in response to the network regulation description, and to configure a gatekeeper interface circuit in response to the network regulation description wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone and wherein the gatekeeper interface circuit is structured to regulate communications between end points of the first network zone with at least one of an external communications portal or an external device.
Claim 2. The system of claim 1, wherein the network regulation description comprises a data request permission description comprising data values	Claim 2. The system of claim 1, wherein the network regulation description comprises a data request permission description comprising data values associated with data

associated with data requestors, wherein at least a portion of the data requestors comprise end points of at least one of the first network zone or the second network zone, and wherein the first network interface circuit is further structured to regulate communications in response to the data request permission description.	requestors, wherein at least a portion of the data requestors comprise end points of at least one of the first network zone or the second network zone, and wherein the first network interface circuit is further structured to regulate communications in response to the data request permission description.
Claim 3. The system of claim 2, wherein: the configuration circuit is further structured to configure a second network interface circuit in response to the network regulation description; and the second network interface circuit is structured to regulate communications of the end points of the second network zone.	Claim 3. The system of claim 2, further comprising: wherein the configuration circuit is further structured to configure a second network interface circuit in response to the network regulation description; and wherein the second network interface circuit is structured to regulate communications of the end points of the second network zone.
Claim 4. The system of claim 3, wherein: the configuration circuit is further structured to configure a gatekeeper interface circuit in response to the network regulation description; and the gatekeeper interface circuit is structured to regulate communications between end points of both the first network zone and the second network zone with an external device.	Claim 4. The system of claim 3, wherein the gatekeeper interface circuit is further structured to regulate communications between end points the second network zone with the at least one of the external communications portal or the external device.
Claim 5. The system of claim 4, wherein the external device is communicatively coupled to one of the first network zone or the second network zone.	Claim 5. The system of claim 4, wherein at least one of the external communications portal or the external device is communicatively coupled to one of the first network zone or the second network zone.
Claim 6. The system of claim 4, wherein the external device is communicatively coupled to a transceiver of the vehicle, wherein the transceiver is communicatively coupled to the second network zone.	Claim 6. The system of claim 4, wherein the external communications portal comprises a transceiver of the vehicle,

	wherein the transceiver is communicatively coupled to the second network zone.
Claim 7. The system of claim 4, wherein the external device comprises an application comprising at least one of: a cloud server-based application, a web based application, or a mobile device application.	Claim 7. The system of claim 4, wherein the external device comprises an application comprising at least one of: a cloud server based application, a web based application, or a mobile device application.
Claim 8. The system of claim 4, wherein the data request permission description further comprises data access permissions associated with at least one of: the external device; a flow associated with one of the end points of the first network zone and/or the second network zone; a flow associated with the external device; a vehicle function associated with one of the end points of the first network zone and/or the second network zone; or a vehicle function associated with the external device; wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to the data access permissions.	Claim 8. The system of claim 4, further comprising: wherein the data request permission description further comprises data access permissions associated with at least one of: at least one of the external communications portal or the external device; a flow associated with one of the end points of the first network zone and/or the second network zone; a flow associated with the external device; a service group associated with one of the end points of the first network zone and/or the second network zone; a service group associated with the external device; a vehicle function associated with one of the end points of the first network zone and/or the second network zone; or a vehicle function associated with the external device; wherein the gatekeeper interface circuit is further structured to regulate communications with the at least one of the external communications portal or the external device in response to the data access permissions.
Claim 9. The system of claim 4, wherein: the data request permission description further comprises an external communication bandwidth limitation; and the gatekeeper interface circuit is further structured to regulate	Claim 9. The system of claim 4, further comprising: wherein the data request permission description further comprises an external communication bandwidth limitation; and wherein the gatekeeper interface circuit is further structured

communications with the external device in response to the external communication bandwidth limitation.	to regulate communications with the at least one of the external communications portal or the external device in response to the external communication bandwidth limitation.
Claim 10. The system of claim 9, wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to at least one of: a flow associated with one of the regulated communications; a data type associated with one of the regulated communications; a data service provider associated with one of the regulated communications; a vehicle function associated with one of the regulated communications; or a connection type of a communicative coupling with the external device.	Claim 10. The system of claim 9, wherein the gatekeeper interface circuit is further structured to regulate communications with the at least one of the external communications portal or the external device in response to at least one of: a flow associated with one of the regulated communications; a data type associated with one of the regulated communications; a data service provider associated with one of the regulated communications; a vehicle function associated with one of the regulated communications; a service group associated with one of the regulated communications; or a connection type of the external communications portal.
Claim 11. The system of claim 1, wherein the configuration circuit is further structured to receive a policy update comprising a change to the network regulation description, and to update the configuration of the first network interface circuit in response to the change to the network regulation description.	Claim 11 The system of claim 1, wherein the configuration circuit is further structured to receive a policy update comprising a change to the network regulation description, and to update the configuration of at least one of the first network interface circuit or the gatekeeper interface circuit in response to the change to the network regulation description.
Claim 12. The system of claim 11, wherein the policy manager circuit is further structured to interpret an authorization associated with the policy update, and to suppress the policy update in response to the authorization indicating a requesting unit is not	Claim 12. The system of claim 11, wherein the policy manager circuit is further structured to interpret an authorization associated with the policy update, and to suppress the policy update in response to the authorization indicating a requesting unit is not authorized to make the

authorized to make the change to the network regulation description of the policy update.	change to the network regulation description of the policy update.
Claim 13. The system of claim 12, wherein the requesting unit comprises at least one unit selected from the units consisting of: **an entity associated with the policy update; an application associated with the policy update**; **a flow associated with the policy update;** a vehicle **function associated with the policy update; and an identifier of an external device communicating the policy update**.	Claim 13. The system of claim 12, wherein the requesting unit comprises at least one unit selected from the units consisting of: an entity associated with the policy update; an application associated with the policy update; a flow associated with the policy update; a mobile system function associated with the policy update; and an identifier of a second external device communicating the policy update.

6. Claims 1-22 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-24 of U.S. Patent No. US 12095677 hereinafter '677. Although the claims at issue are not identical, they are not patentably distinct from each other because each element of the claims of the instant application is anticipated claim 1 of patent '677.

See, the table below which shows both application claims on limitation bases.

The instant application - 18/904,768	**Patent No. US 12095677**
Claim 1. A system, comprising:	Claim 1. **A system, comprising:**

a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description; a configuration circuit structured to configure a first network interface circuit in response to the network regulation description; and wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.	a mobile system having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description; a configuration circuit structured to configure a first network interface circuit in response to the network regulation description, and to configure a gatekeeper interface circuit in response to the network regulation description; wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone; and wherein the gatekeeper interface circuit is structured to regulate communications between end points of the first network zone with at least one of an external communications portal or an external device.
Claim 2. The system of claim 1, wherein the network regulation description comprises a data request permission description comprising data values associated with data requestors, wherein at least a portion of the data requestors comprise end points of at least one of the first	Claim 2. The system of claim 1, wherein the network regulation description comprises a data request permission description comprising data values associated with end points of at least one of the first network zone or the second network zone, and wherein the first network

network zone or the second network zone, and wherein the first network interface circuit is further structured to regulate communications in response to the data request permission description.	interface circuit is further structured to regulate communications in response to the data request permission description.
Claim 4. The system of claim 3, wherein: the configuration circuit is further structured to configure a gatekeeper interface circuit in response to the network regulation description; and the gatekeeper interface circuit is structured to regulate communications between end points of both the first network zone and the second network zone with an external device.	Claim 4. The system of claim 3, wherein the gatekeeper interface circuit is further structured to regulate communications between the second network zone and the at least one of the external communications portal or the external device.
Claim 5. The system of claim 4, wherein the external device is communicatively coupled to one of the first network zone or the second network zone.	Claim 5. The system of claim 4, wherein the external communications portal or the external device is communicatively coupled to the first network zone or the second network zone.
Claim 6. The system of claim 4, wherein the external device is communicatively coupled to a transceiver of the vehicle, wherein the transceiver is communicatively coupled to the second network zone.	Claim 6. The system of claim 4, wherein the external communications portal comprises a transceiver of the mobile system, wherein the transceiver is communicatively coupled to the second network zone.
Claim 7. The system of claim 4, wherein the external device comprises an application comprising at least one of: a cloud server-based application, a web based application, or a mobile device application.	Claim 7. The system of claim 4, wherein the external device comprises at least one of: a cloud server based application, a web based application, or a mobile device application.
Claim 8. The system of claim 4, wherein the data request permission description further comprises data access permissions associated with at least one of:	Claim 8. The system of claim 4, further comprising: wherein the data request permission description further comprises data access permissions associated with at

the external device; a flow associated with one of the end points of the first network zone and/or the second network zone; a flow associated with the external device; a vehicle function associated with one of the end points of the first network zone and/or the second network zone; or a vehicle function associated with the external device; wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to the data access permissions.	least one of: at least one of the external communications portal or the external device; a flow associated with one of the end points of the first network zone and/or the second network zone; a flow associated with the external device; a service group associated with one of the end points of the first network zone and/or the second network zone; a service group associated with the external device; a mobile system function associated with one of the end points of the first network zone and/or the second network zone; or a mobile system function associated with the external device; wherein the gatekeeper interface circuit is further structured to regulate communications with the at least one of the external communications portal or the external device in response to the data access permissions.
Claim 9. The system of claim 4, wherein: the data request permission description further comprises an external communication bandwidth limitation; and the gatekeeper interface circuit is further structured to regulate communications with the external device in response to the external communication bandwidth limitation.	Claim 9. The system of claim 4, further comprising: wherein the gatekeeper interface circuit is further structured to regulate communications with the at least one of the external communications portal or the external device in response to an external communication bandwidth limitation of the data request permission description.
Claim 10. The system of claim 9, wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to at least one of: a flow associated with one of the regulated communications;	Claim 10. The system of claim 9, wherein the gatekeeper interface circuit is further structured to regulate communications with the at least one of the external communications portal or the external device in response to at least one of: a flow associated with one of the regulated communications; a data type associated with one of the

a data type associated with one of the regulated communications; a data service provider associated with one of the regulated communications; a vehicle function associated with one of the regulated communications; or a connection type of a communicative coupling with the external device.	regulated communications; a data service provider associated with one of the regulated communications; a mobile system function associated with one of the regulated communications; a service group associated with one of the regulated communications; or a connection type of the external communications portal.
Claim 11. **The system of claim 1, wherein the configuration circuit is further structured to receive a policy update comprising a change to the network regulation description, and to update the configuration of the first network interface circuit in response to the change to the network regulation description.**	Claim 11. The system of claim 1, wherein the configuration circuit is further structured to receive a policy update comprising a change to the network regulation description, and to update the configuration of at least one of the first network interface circuit or the gatekeeper interface circuit in response to the change to the network regulation description.
Claim 12. The system of claim 11, wherein the policy manager circuit is further structured to interpret an authorization associated with the policy update, and to suppress the policy update in response to the authorization indicating a requesting unit is not authorized to make the change to the network regulation description of the policy update.	Claim 12. The system of claim 11, wherein the policy manager circuit is further structured to interpret an authorization associated with the policy update, and to suppress the policy update in response to the authorization indicating a requesting unit is not authorized to make the change to the network regulation description of the policy update.
Claim 13. The system of claim 12, wherein the requesting unit comprises at least one unit selected from the units consisting of: **an entity associated with the policy update; an application associated with the policy update; a flow associated with the policy update;**	Claim 13. The system of claim 12, wherein the requesting unit comprises at least one unit selected from the units consisting of: an entity associated with the policy update; an application associated with the policy update; a flow associated with the policy update; a mobile system function associated with the policy update; and an

a vehicle **function associated with the policy update; and an identifier of an external device communicating the policy update**.	**identifier of a second external device communicating the policy update.**
Claim 14. A method, comprising: interpreting a policy comprising a network regulation description; **configuring a first network interface circuit in response to the network regulation description**; and **regulating communications, using the first network interface circuit, between end points of a first network zone of a vehicle and end points of a second network zone of the vehicle.**	**Claim 15. A method, comprising: interpreting a policy comprising a network regulation description;** **configuring a first network interface circuit in response to the network regulation description;** configuring a gatekeeper interface circuit in response to the network regulation description; **regulating communications, using the first network interface circuit, between end points of a first network zone of a vehicle** and end points of a second network zone of the vehicle; and further regulating communications, using the gatekeeper interface circuit, between end points of the first network zone of the vehicle and at least one of an external communications portal or an external device.
Claim 15. The method of claim 14, wherein the network regulation description comprises a data request permission description comprising data values associated with data requestors, wherein at least a portion of the data requestors comprise end points of at least one of **the first network zone or the second network zone**; and the method further comprises: **regulating communications in response to the data request permission description.**	Claim 16. **The method of claim 15**, further comprising: **wherein the network regulation description comprises a data request permission description comprising data values associated with data requestors, wherein at least a portion of the data requestors comprise: end points of the first network zone**; end points of **the second network zone**; or the external device; and **further regulating communications in response to the data request permission description.**

Claim 16. The method of claim 15, further comprising: configuring a second network interface circuit in response to the network regulation description; and regulating communications, using the second network interface circuit, of the end points of the second network zone.	**Claim 17.** The method of claim 16, further comprising: configuring a second network interface circuit in response to the network regulation description; and regulating communications, using the second network interface circuit, of the end points of the second network zone.
Claim 17. The method of claim 16, further comprising: configuring a gatekeeper interface circuit in response to the network regulation description; and regulating communications, using the gatekeeper interface circuit, between end points of both the first network zone and the second network zone with an external device.	Claim 18. The method of claim 17, further comprising: configuring a gatekeeper interface circuit in response to the network regulation description; and further regulating communications, using the gatekeeper interface circuit, between end points of the second network zone with the at least one of the external communications portal or the external device.

7. Claims 1-22 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-29 of U.S. Patent No. US 12244464 hereinafter '464. Although the claims at issue are not identical, they are not patentably distinct from each other because each element of the claims of the instant application is anticipated claim 1 of patent '464.

Please see the table below which shows both application claims on limitation bases.

The instant application - 18/904,768	Patent No. US 12244464
Claim 1. A system, comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone;	Claim 1. A system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone;

a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description; a configuration circuit structured to configure a first network interface circuit in response to the network regulation description; and wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.	a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description and an external communication permission value; a configuration circuit structured to configure at least one network interface circuit in response to the policy and to configure a gatekeeper interface circuit in response to the policy; the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone; and the gatekeeper interface circuit, wherein the gatekeeper interface circuit is configured to regulate communications between: end points of at least one of the first network zone or the second network zone; and an external device.
Claim 14. A method, comprising: interpreting a policy comprising a network regulation description; configuring a first network interface circuit in response to the network regulation description; and regulating communications, using the first network interface circuit, between end points of a first network	Claim 9. A method, comprising: interpreting a policy comprising a network regulation description and an external communication permission value; regulating, in response to the network regulation description, communications between end points of a first network zone of a vehicle and end points of a second

zone of a vehicle and end points of a second network zone of the vehicle.	**network zone of the vehicle** via performing one of an up-sampling operation or a down-sampling operation on the communication regulating communications, in response to the external communication permission value, between: end points of at least one of the first network zone or the second network zone; and an external device; and wherein the second network zone is of a different type than the first network zone.

8. Claims 1-22 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-29 of co-pending 18/908,478 hereinafter ‘478. Although the claims at issue are not identical, they are not patentably distinct from each other because the instant claims and the Co-pending ‘478 claims share the same scope with minimal differences that would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify the instant claims in a way to match the co-pending claims limitation. The instant claims discloses a converged network device (CND) interposed between the first network zone and the second network zone. CND reduced infrastructure complexity, lower capital and operational costs, and improved performance. It combines data and storage networking, simplifying management and reducing cables, power usage, and latency. In addition, the instant claims use the term "a network regulation description", wherein the co-pending claims use the term "an external data routing description and an external data service description". However, these terms are claimed in a broad and general terms without

claiming any specific details that distinguish between them and thus "an external data routing description and an external data service description" can be an example of "a network regulation description".

Similar reasoning applies to claim 14 of the instant application. See, the table below which shows both application claims on limitation bases.

The instant application - 18/904,768	**Co-pending 18/908,478**
Claim 1. A system, comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description; a configuration circuit structured to configure a first network interface circuit in response to the network regulation description; and wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.	Claim 1. **A system, comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone;** **a policy manager circuit structured to interpret a policy comprising** an external data routing **description** and an external data service description; **a configuration circuit structured to configure** a gatekeeper **interface circuit in response to** the external data routing description and the external data service description; the gatekeeper interface circuit interposed between the first network zone and at least one external communication portal selectively couplable to an external device, and further interposed

	between the second network zone and the at least one external communication portal; and wherein the gatekeeper **interface circuit is structured to regulate communications between end points of the first network zone and** the at least one external communication portal, and to regulate communications between **end points of the second network zone** and the at least one external communication portal.

9. Claims 1-22 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-29 of co-pending 18/908,498 hereinafter '498. Although the claims at issue are not identical, they are not patentably distinct from each other because each element of the claims of the instant application is anticipated claim 1 of patent '498.

Similar reasoning applies to claim 14 of the instant application. See, the table below which shows both application claims on limitation bases.

The instant application - 18/904,768	**Co-pending 18/908,498**
Claim 1. A system, comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone;	Claim 1. **A system, comprising:** **a vehicle having** at least one **network zone;** **a converged network device (CND) interposed between the** at least one **network zone** and at least one of an external

a **converged network device (CND) interposed between the** first **network zone** and the second network zone, the CND comprising: **a policy manager circuit structured to interpret a policy comprising a network regulation description**; **a configuration circuit structured to configure** a first network **interface circuit in response to** the network regulation description; and **wherein the** first network **interface circuit is structured to regulate communications between end points of the first network zone** and end points of the second network zone.	communication portal or a first external device, **the CND comprising:** **a policy manager circuit structured to interpret a policy comprising a network regulation description**, wherein the policy is communicated to the CND from a second external device; and **a configuration circuit structured to configure** a gatekeeper **interface circuit in response to** the policy; and **wherein the** gatekeeper **interface circuit is structured to regulate communications between end points of the** at least one **network zone and** the at least one of the external communication portal or the first external device.

Claim Rejections - 35 USC § 103

10. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such

that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

11. Claims 1, 11, 13-14 and 21 are rejected under 35 U.S.C. 103 as being unpatentable over Han et al. (US 20180076970 hereinafter Han) in view of Richmond et al. (US 20200274927 hereinafter Richmond).

12. **With respect to claims 1 and 14,** Han teaches a system, comprising:

a vehicle having a first network zone and a second network zone of a different type than the first network zone (Han, see paragraph [0008] an operation method of a **first communication node supporting communications between an Ethernet-based network** and **a controller area network (CAN) may comprise receiving an Ethernet message from a second communication node belonging to the Ethernet-based network (i.e. CAN is the first network zone and "Ethernet-based network" is the second network zone in a vehicle and CAN and Ethernet-based network are different type networks)**; performing an integrity verification on first automotive safety integrity level (ASIL) authentication information included in the Ethernet message; generating a CAN message based on the Ethernet message for which the integrity verification has been completed);

a converged network device (CND) interposed between the first network zone and the second network zone (Han, see paragraph [0029] a **gateway that supports communications between the CAN and the Ethernet-based network** can receive the message from the end node (i.e. gateway is the CND interposed between CAN and

Ethernet- based network), and **perform an integrity verification operation on the message based on the ASIL authentication information**. When the integrity verification of the message is successfully completed, the gateway can generate a new message containing an integrity-verified data unit, and transmit the generated message to the CAN or Ethernet-based network), the CND comprising:

wherein the first network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone(Han, see paragraph [0130] If the destination of the CAN message is a communication node belonging to the Ethernet-based network, the gateway 200 may generate an Ethernet message (e.g., the Ethernet message 300 shown in FIGS. 3 and 4) based on the corresponding CAN message (S1605)).

Han yet fails to explicitly disclose , the CND comprising:

a policy manager circuit structured to interpret a policy comprising a network regulation description;

a configuration circuit structured to configure a first network interface circuit in response to the network regulation description;

However, Richmond discloses the CND comprising:

a policy manager circuit structured to interpret a policy comprising a network regulation description (Richmond, see paragraph [0039, 0043] network description files 1 to N can be defined for different networks or a mixed network, which can include CAN, LIN or Ethernet networks operating within vehicle 110. For other examples, network description files can be defined for the same network and protocol. At block 305, the

network description files 1 to N are converted into a uniform network description file 307 (i.e., according to policy network regulation description). For example, server 207 or, alternatively, vehicle gateway 127 can receive network descriptor files 1 to N and convert the files into a uniform network description file as structured in FIG. 3B…);

a configuration circuit structured to configure a first network interface circuit in response to the network regulation description (Richmond, see paragraph [0046] a process 450 of **auto-generating updated ECU files or code**. At block 417, **updated DBC files for ECU 1 (412-1) through ECU N (412-N) are input into a translator. Translator 417 translates the updated DBC files into an updated centralized connectivity description into a hierarchical format describing the updated (i.e., configuration)** connectivity within a network topology of a vehicle (i.e., network regulation descriptions, allowing for updates to security rules, communication protocols, and the configuration circuit translates these policies into specific actions for gatekeeper interfaces)… Claim 2 further discloses the interface is to **receive one or more updated network descriptor files and the processor configured to convert the one or more updated network descriptor files into an updated uniform network descriptor file**).

Therefore, it would have been obvious to one of ordinary skill in the art at the time the invention was effectively filed to combine the teaching of Han with the teaching of Richmond to provide the system for a policy manager and a configuration circuit in a vehicle system provides an efficient, flexible, and secure approach to managing complex, heterogeneous in-vehicle networks. This architecture allows for the dynamic, automated regulation of network traffic and communication between different network zones, such as powertrain, body electronics, and telematics.

13. **With respect to claims 11 and 21,** Han-Richmond teaches the system**,** wherein the configuration circuit is further structured to receive a policy update comprising a change to the network regulation description, and to update the : configuration of the first network interface circuit in response to the change to the network regulation description(Richmond, see paragraph [0046] a process 450 of **auto-generating updated ECU files or code**. At block 417, **updated DBC files for ECU 1 (412-1) through ECU N (412-N) are input into a translator. Translator 417 translates the updated DBC files into an updated centralized connectivity description into a hierarchical format describing the updated (i.e., configuration)** connectivity within a network topology of a vehicle (i.e., network regulation descriptions, allowing for updates to security rules, communication protocols, and the configuration circuit translates these policies into specific actions for gatekeeper interfaces).

14. **With respect to claim 13,** Han-Richmond teaches the system, wherein the requesting unit comprises at least one unit selected from the units consisting of:

an entity associated with the policy update(Richmond, see paragraph [0046] a process 450 of **auto-generating updated ECU files or code**. At block 417, **updated DBC files for ECU 1 (412-1) through ECU N (412-N) are input into a translator. Translator 417 translates the updated DBC files into an updated centralized connectivity description into a hierarchical format describing the updated (i.e., configuration)** connectivity within a network topology of a vehicle (i.e., network regulation descriptions, allowing for updates to security rules, communication protocols);

an application associated with the policy update;

a flow associated with the policy update; a vehicle function associated with the policy update; and an identifier of an external device communicating the policy update.

15. Claims 2-4, 9-10, 12, 15-17, 19-20 and 22 are rejected under 35 U.S.C. 103 as being unpatentable over Han et al. (US 20180076970 hereinafter Han) in view of Richmond et al. (US 20200274927 hereinafter Richmond) further in view of Barrett et al. (US 20190394089 hereinafter Barrett).

16. **With respect to claims 2 and 15,** Han-Richmond teaches the system, yet fails to explicitly disclose wherein the network regulation description comprises a data request permission description comprising data values associated with data requestors, wherein at least a portion of the data requestors comprise end points of at least one of the first network zone or the second network zone, and wherein the first network interface circuit is further structured to regulate communications in response to the data request permission description.

However, Barrett discloses wherein the network regulation description comprises a data request permission description comprising data values associated with data requestors, wherein at least a portion of the data requestors comprise end points of at least one of the first network zone or the second network zone, and wherein the first network interface circuit is further structured to regulate communications in response to the data request permission description (Barrett, see paragraphs [0034, 0037] the firewall system can include one or more firewalls that can filter inbound and outbound

network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow **packets of a type that appear on a whitelist to pass through the firewall (i.e., permission)**, ...).

Therefore, it would have been obvious to one of ordinary skill in the art at the time the invention was effectively filed to combine the teaching of Han-Richmond with the teaching of Barrett to provide the system for a network regulation scheme comprising a data request permission description with data values associated with data requestors that is used by a network interface circuit to regulate communication, is the ability to achieve fine-grained, context-aware access control for specific endpoints within network zones. By permissions directly to data values and specific endpoints rather than just broad network locations, the system provides several key benefits such as: improved security posture and data protection, regulatory compliance and enhanced visibility and response by associating requestor data with specific endpoints in the first or second zone.

17. **With respect to claims 3 and 16,** Han-Richmond-Barrett teaches the system, wherein: the configuration circuit is further structured to configure a second network interface circuit in response to the network regulation description; and the second network interface circuit is structured to regulate communications of the end points of the second network zone (Richmond, see paragraph [0007] the network topology of the vehicle includes **Controller Area Network (CAN),** Local Interconnect **Network (LIN), or Ethernet networks (i.e. CAN is the first network zone and "Ethernet-based**

network" is the second network zone). The one or more network descriptor files **or updated network descriptor files includes a CAN network descriptor file or updated CAN network descriptor file**, a LIN network descriptor file or updated LIN descriptor file, or an Ethernet network descriptor file or updated Ethernet network descriptor file. Such network descriptor files for these individual networks can be converted into a uniform network descriptor file or an updated uniform descriptor file that describes the network topology including the individual networks without being specific to any network).

18. **With respect to claims 4 and 17,** Han-Richmond-Barrett teaches the system, wherein: the configuration circuit is further structured to configure a gatekeeper interface circuit in response to the network regulation description; and the gatekeeper interface circuit is structured to regulate communications between end points of both the first network zone and the second network zone with an external device (Barrett, see paragraphs [0034, 0037] the firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, **while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report**, and/or take other action).

19. **With respect to claims 9 and 19,** Han-Richmond-Barrett teaches the system, wherein: the data request permission description further comprises an external communication bandwidth limitation; and the gatekeeper interface circuit is further structured to regulate communications with the external device in response to the external communication bandwidth limitation (Barrett, see paragraph [0019] the described techniques can enhance security and reliability of communications over the vehicle network. The example techniques can reduce the risk of successful cyberattacks and improve driver safety for connected cars technologies. The example techniques can also be used for autonomous driving which requires secure connectivity and hi**gh-bandwidth communications between ECUs across functional domains.** The example techniques can apply to any automotive or vehicular networking technology including, for example, IP/Ethernet, CAN (Control Area Network), LIN (Local Interconnect Network), Flexray, and Media Oriented System Transport (MOST)).

20. **With respect to claims 10 and 20,** Han-Richmond-Barrett teaches the system, wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to at least one of:

a flow associated with one of the regulated communications;

a data type associated with one of the regulated communications;

a data service provider associated with one of the regulated communications;

a vehicle function associated with one of the regulated communications(Richmond, see paragraph [0046] a process 450 of **auto-generating updated ECU files or code**. At

block 417, **updated DBC files for ECU 1 (412-1) through ECU N (412-N) are input into a translator. Translator 417 translates the updated DBC files into an updated centralized connectivity description into a hierarchical format describing the updated (i.e., configuration)** connectivity within a network topology of a vehicle (i.e., network regulation descriptions, allowing for updates to security rules, communication protocols, and the configuration circuit translates these policies into specific actions for gatekeeper interfaces); or a connection type of a communicative coupling with the external device.

21. **With respect to claims 12 and 22,** Han-Richmond teaches the system, yet fails to explicitly disclose wherein the policy manager circuit is further structured to interpret an authorization associated with the policy update, and to suppress the policy update in response to the authorization indicating a requesting unit is not authorized to make the change to the network regulation description of the policy update.

However, Barrett discloses wherein the policy manager circuit is further structured to interpret an authorization associated with the policy update, and to suppress the policy update in response to the authorization indicating a requesting unit is not authorized to make the change to the network regulation description of the policy update(Barrett, see paragraphs [0034, 0037] the firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear

on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action).

Same motivation as claim 2.

22. Claims 5-8 and 18 are rejected under 35 U.S.C. 103 as being unpatentable over Han et al. (US 20180076970 hereinafter Han) in view of Richmond et al. (US 20200274927 hereinafter Richmond) in view of Barrett et al. (US 20190394089 hereinafter Barrett) further in view of Choi (CN 109466471 hereinafter Choi).

23. **With respect to claim 5,** Han-Richmond-Barrett teaches the system, yet fails to explicitly disclose wherein the external device is communicatively coupled to one of the first network zone or the second network zone.

However, Choi discloses wherein the external device is communicatively coupled to one of the first network zone or the second network zone (Choi, see page 7, lines 34-38, controller 200 may determine whether the un-authorized personnel entering the system of the vehicle through vehicle remote information processing or telephone connectivity, **whether through the connection to the external device of the vehicle 1 carried out, or whether implemented through controller area network (CAN) or Ethernet as the vehicle communication. the description its detailed description** referring to FIG. 6).

Therefore, it would have been obvious to one of ordinary skill in the art at the time the invention was effectively filed to combine the teaching of Han-Richmond-Barrett with the teaching of Choi to provide a communicatively coupling an external device to a specific network zone such as a first or second network zone/segment, provides several key advantages, primarily centered on security, performance, and operational flexibility. By placing a device in a specific zone, organizations can control, monitor, and isolate its traffic.

24. **With respect to claim 6,** Han-Richmond-Barrett teaches the system, yet fails to explicitly disclose wherein the external device is communicatively coupled to a transceiver of the vehicle, wherein the transceiver is communicatively coupled to the second network zone.

However, Choi discloses wherein the external device is communicatively coupled to a transceiver of the vehicle, wherein the transceiver is communicatively coupled to the second network zone(Choi, see page 7, lines 34-38, controller 200 may determine whether the un-authorized personnel entering the system of the vehicle through vehicle remote information processing or telephone connectivity, **whether through the connection to the external device of the vehicle 1 carried out, or whether implemented through controller area network (CAN) or Ethernet as the vehicle communication. the description its detailed description** referring to FIG. 6).

Same motivation as claim 5.

25. **With respect to claim 7,** Han-Richmond-Barrett teaches the system, yet fails to explicitly disclose wherein the external device comprises an application comprising at least one of: a cloud server-based application, a web based application, or a mobile device application.

However, Choi discloses wherein the external device comprises an application comprising at least one of: a cloud server-based application, a web based application, or a mobile device application (Choi, see FIG. 4 and page 8, lines 10-15, when determining whether to implement an intrusion refers to wireless network can determine whether the network area (item 1), the **remote information processing area (item 2) and (wireless updating** firmware over the air; (FOTA) region (item 3) is subjected to invasion, and when connected to the external device of the vehicle 1 exists, it can be **determined whether to implement intrusion in connection to the vehicle 1 of the external device**, a universal serial bus (USB), SD card or **intelligent mobile phone).**

Therefore, it would have been obvious to one of ordinary skill in the art at the time the invention was effectively filed to combine the teaching of Han-Richmond-Barrett with the teaching of Choi to provide an external device such as a sensor, smart hardware, or wearable, utilizing a cloud-server-based, web-based, or mobile application is that it offloads heavy processing, storage, and UI management to the cloud or a smarter device, enabling the external device to remain lightweight, energy-efficient, and cost-effective while providing a superior, synchronized user experience.

With respect to claims 8 and 18, Han-Richmond-Barrett teaches the system**,** yet fails to explicitly disclose wherein the data request permission description further comprises data access permissions associated with at least one of:

the external device; a flow associated with one of the end points of the first network zone and/or the second network zone; a flow associated with the external device; a vehicle function associated with one of the end points of the first network zone and/or the second network zone; or a vehicle function associated with the external device; wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to the data access permissions.

However, Choi discloses wherein the data request permission description further comprises data access permissions associated with at least one of:

the external device (Choi, see page 7, lines 34-38, whether through the connection to the external device of the vehicle 1 carried out, or whether implemented through controller area network (CAN) or Ethernet as the vehicle communication);

a flow associated with one of the end points of the first network zone and/or the second network zone;

a flow associated with the external device;

a vehicle function associated with one of the end points of the first network zone and/or

the second network zone; or a vehicle function associated with the external device;

wherein the gatekeeper interface circuit is further structured to regulate communications with the external device in response to the data access permissions.

Same motivation as claim 5.

Conclusion

26. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. This includes:

PG. Pub. US 20190356574 Motor vehicle e.g. passenger car has domain controllers which are provided to transmit one of messages between two of separate security zones and to block transmission of message if security criterion is violated.

PG. Pub. US 20170093659 Controller area network device e.g. controller area network transceiver integrated circuit device, has signal generator for outputting signal to invalidate controller area network message in response to match signal.

PG. Pub. US 20090187968 Network e.g. wide area network, service usage controlling method for network system, involves storing portion of saved set and modified policies on forwarding devices, and storing remainder of policies on infrastructure.

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. See PTO-892 Notice of References Cited.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ELIZABETH KASSA whose telephone number is (571)270-0567. The examiner can normally be reached on Monday -Friday 9 AM -6 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an

interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ario Etienne can be reached on 517-272-4001. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppair-my.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

03/17/2020

/ELIZABETH KASSA/
Examiner, Art Unit 2457

/ARIO ETIENNE/
Supervisory Patent Examiner, Art Unit 2457